



Enhancing droplet transfer in laser-arc hybrid welding via oscillating laser-induced plasma interaction

Yuxi Ma^{a,*}, Jiaqi Zhang^a, Hao Xu^a, Shi Qiao^a, Ming Gao^b, Lei Wang^{c,**}

^a School of Civil Engineering and Architecture, Wuhan Polytechnic University, Wuhan, 430023, PR China

^b Wuhan National Laboratory for Optoelectronics (WNLO), Huazhong University of Science and Technology, Wuhan, 430074, PR China

^c SZJ Automation Co., Ltd, Shenzhen, 518105, PR China

ARTICLE INFO

Keywords:

Aluminum alloy
Laser beam oscillating
Laser-arc hybrid welding
Droplet transfer
Plasma interaction

ABSTRACT

Controlling droplet transfer is crucial for achieving high-quality laser-arc hybrid welding of aluminum alloys. This study systematically investigates the effects of oscillating laser beam modes (transverse, longitudinal, circular) on the welding process of 8 mm thick AA6082 aluminum alloy. High-speed imaging reveals that non-scanning mode leads to coarse projected transfer with significant spatter. While transverse and longitudinal scanning improve stability, they still involve medium projected and short-circuiting transfer. In contrast, circular scanning uniquely promotes stable fine projected transfer, reducing the average droplet diameter by 47% (to 0.9 mm) and transfer time by 44% (to 3.8 ms), while drastically suppressing spatter. Plasma spectroscopy analysis provides the underlying physical mechanism: circular scanning most effectively enhances laser-arc plasma interaction, increasing arc zone electron temperature by ~8.5% and electron density by 43%. This reduces arc resistivity, promotes wire melting, refines the molten metal column necking diameter ($D \sim R \sim$ to 0.45 mm), and stabilizes the arc force direction. Consequently, the critical wavelength for liquid column instability decreases, facilitating droplet detachment. This work elucidates the mechanism of droplet regulation by oscillating lasers and identifies circular scanning as the optimal strategy for high-quality aluminum alloy hybrid welding.

1. Introduction

Aluminum alloys have become core structural materials in aerospace, new energy vehicles, and other fields due to their excellent specific strength, corrosion resistance, and lightweight characteristics [1–5]. However, aluminum alloys have properties such as high thermal conductivity, ease of oxidation, and high reflectivity, placing high demands on welding materials, welding speed, and heat input when using arc welding. Laser welding, on the other hand, faces challenges such as low absorption rates, high porosity, and high costs [6,7].

Steen.W proposed the concept of laser-arc hybrid welding in 1980 [8, 9]. This technology takes full advantage of the benefits of both arc welding and laser welding, and the synergistic effect produced by the interaction of the arc and laser reduces the limitations of using each method alone. The high bridging capability of arc welding, combined with the high energy density of laser welding, significantly improves welding efficiency and quality. This reduces the number of welding passes required to achieve the same penetration depth, lowers heat

input, minimizes the heat-affected zone, and reduces porosity and cracking defects, making it an important direction for aluminum alloy welding.

Research on laser-arc hybrid welding has focused on numerous welding parameters, such as the selection and flow rate of shielding gases, laser power and focal position, and arc angle. Xu et al. prepared a high-performance, low-cost titanium alloy by controlling the flow rate of shielding gas [10]. Sun et al. studied the effect of laser power on weld characteristics and microstructure by welding a steel used in marine environments [11]. Feng et al. investigated the effect of different laser-arc separation distances on droplet transfer stability, weld pool flow, and weld quality when welding AH36 steel [12].

The introduction of oscillating laser technology provides a new approach for dynamically controlling the coupling behavior of hybrid heat sources [13]. By adjusting scanning modes (e.g., transverse, radial, circular) and parameters (amplitude, frequency), this technology can control plasma distribution and droplet transfer during welding, ultimately improving weld formation.

* Corresponding author.

** Corresponding author.

E-mail addresses: mayuxi@whpu.edu.cn (Y. Ma), lbw984@126.com (L. Wang).

Wang et al. found that circular scanning is the most favorable laser scanning method for welding stability and weld formation during hybrid oscillating welding of AA6061-T6 aluminum alloy [14]. Miao et al. studied the surface formation law of aluminum alloy welds under different scanning parameters, provided the optimal range of scanning amplitude and scanning frequency for high-frequency oscillating hybrid welding, and established a prediction model for weld morphology characteristics [15]. Ke et al. studied the mechanism of pore formation caused by keyhole collapse under circular oscillating [16]. However, a systematic and in-depth exploration of how different oscillating modes affect droplet transfer behavior and the associated plasma interaction mechanisms during aluminum alloy laser-arc hybrid welding is still lacking.

Recent investigations have highlighted the critical influence of scanning strategies and plasma-arc interactions on droplet transfer stability in hybrid welding processes [17–19], motivating the present study on oscillating laser-induced plasma effects. Therefore, this paper focuses on 8 mm thick AA6082 aluminum alloy welded with ER5087 filler wire. Utilizing a high-frequency oscillating laser-arc hybrid welding platform combined with high-speed imaging and plasma spectroscopy, we investigate the influence laws of scanning modes on droplet transfer behavior (mode, diameter, time) and plasma characteristics. By uncovering the physical mechanisms through which oscillating laser beams, especially circular scanning, reduce arc resistivity and promote droplet refinement, we aim to provide a deeper understanding of weld formation phenomena and a theoretical basis for high-quality aluminum alloy welding.

2. Material and method

2.1. Material preparation

The base material selected was 8 mm thick AA6082 aluminum alloy plate, and the welding wire selected was 1.2 mm diameter ER5087 aluminum alloy welding wire. Their chemical compositions are shown in Table 1. Before welding, the oxide film on the surface of the base material was first removed by mechanical grinding, and then the oil was removed by wiping with acetone solution. The base material and welding wire were then placed in a drying cabinet for later use. The high-frequency oscillating laser-arc hybrid welding test platform used was mainly composed of a 6000W fiber laser, a 6-axis industrial robot, a pulsed arc welding machine, and an oscillating laser welding head. The oscillating laser welding head can drive the laser beam to oscillate and scan along any trajectory through a scanning galvanometer, and its collimation-focusing focal length ratio is 0.8.

2.2. Test and analysis methods

The test was carried out in the form of bead-on-plate welding. Three laser beam scanning modes (Fig. 1, transverse, radial, and circular) were adopted. A single-factor method was used for the test. The length of each weld was 120 mm, and each parameter was repeated 3 times to ensure the accuracy of the results. The selection range of scanning amplitude was 0.2–2.5 mm, and the selection range of scanning frequency was 10–500 Hz.

Unless otherwise specified, the following process parameters were fixed in the test: laser power $P = 5000\text{W}$, welding current $I = 200\text{A}$, welding speed $v_w = 2\text{ m/min}$, defocusing amount $\Delta f = 0\text{ mm}$, shielding gas flow rate 20 L/min , and laser-arc separation distance $D_{LA} = 3\text{ mm}$.

Table 1

Chemical composition of welding materials (wt%).

Composition	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Zr	Al
6082-T6 Board	1.0	0.50	0.10	0.7	0.9	0.25	0.20	0.10	/	Balance
ER5087 Wire	0.04	0.14	0.01	0.75	4.76	0.007	0.01	0.05	0.11	

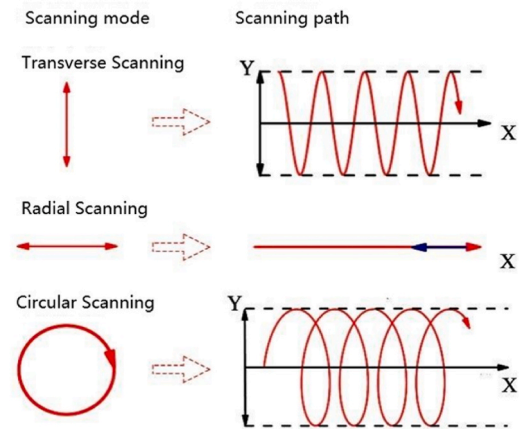


Fig. 1. Laser Scanning modes and paths.

The representative oscillating scanning parameters of $A = 0.6\text{ mm}$ and $f = 300\text{ Hz}$ were selected because they provided stable laser-arc coupling conditions and enabled a clear comparison among different oscillating scanning modes, while avoiding excessive process instability. High-speed imaging was used to capture the welding process to observe droplet transfer characteristics, weld pool flow state, and pulsed arc plasma plume morphology. The high-speed photography equipment used in this study mainly includes a Phantom V710 high-speed camera, a Cavitax HF background light source, a TOKINA macro lens, and filters. The main shooting parameters are: frame rate 7000 fps, exposure time $1\text{ }\mu\text{s}$, laser wavelength of the background light source 810 nm , and filter bandpass $810 \pm 5\text{ nm}$ (Fig. 2). After welding, a digital camera was used to capture the surface morphology of the weld bead. A 100 mm long section in the middle of the weld was selected for surface defect statistics. Typical unscanned and scanned weld surface morphologies are shown in Fig. 3. Here, spatter with a diameter less than 0.3 mm is defined as fine spatter, and that with a diameter greater than 0.3 mm is defined as coarse spatter. The width of the spatter zone formed by fine spatter and the number of coarse spatter particles were counted. The compactness of the fish-scale texture is characterized by the change in the angle of the fish-scale pattern.

A spectral signal acquisition system was used to collect spectral signals during the welding process. This system mainly consists of a plasma spectrometer, an optical transmission system, and a signal triggering system. The spectral signal acquisition platform and collection positions are shown in Fig. 4. During welding, the light collector is fixed on the robot arm and moves synchronously with it. When reaching the specified position, the signal trigger sends a collection signal to the

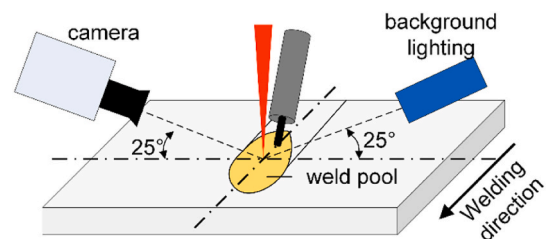


Fig. 2. Schematic diagram of the high-speed photography shooting process.

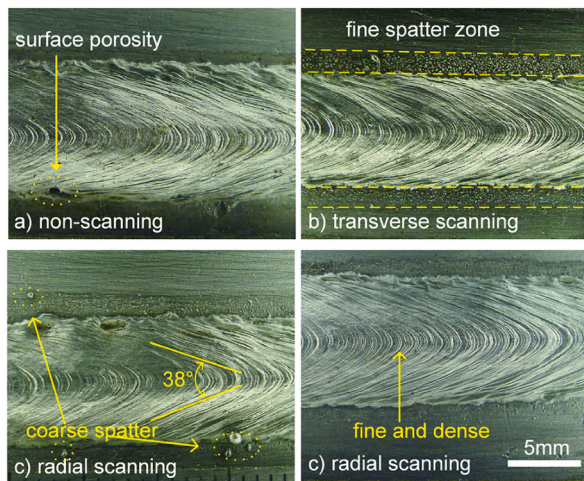


Fig. 3. Typical non-scanned and scanned weld surface morphology.

spectrometer, which then begins to acquire spectral signals. In this study, the spectral collection positions are shown in Fig. 4: one is the laser irradiated zone (LaserZ) 2 mm above the base material, and the other is the central plasma zone (ArcZ). The selected spectrometer is an Advantest four-channel spectrometer with a wavelength resolution range of 200–825 nm, a spectral acquisition interval of 1.1 ms to 1 min, an average resolution of 0.22 nm, a data transfer speed of 1.8 ms per reading, and a signal-to-noise ratio of 200:1. To maximize measurement accuracy, the spectral acquisition interval in this study was set to 1.1 ms, with an acquisition time of 1 s.

By analyzing the typical spectra under non-scanning conditions (Fig. 5), it was found that the Mg II 279.76 nm ion line is clearly distinguishable during both the arc peak and baseline stages. Due to the higher Mg content in the welding wire (4.76 wt%) compared to the base material (0.8 wt%), the Mg II ions detected in the LaserZ zone primarily originate from the transport in the ArcZ zone. Therefore, the intensity of this spectral line can serve as an effective indicator for characterizing the strength of laser-arc interaction.

Based on the collected spectral data, the plasma electron temperature (T_e) was calculated using the Boltzmann plot method (Formula 1), with multiple Mg I atomic lines (such as 518.44 nm, 552.96 nm) selected for fitting. The electron density (n_e) was calculated using the Stark broadening method, selecting the Mg I 552.96 nm line with a larger half-width, and the value was obtained using Formula 2. The reported electron temperature and electron density values were obtained by

averaging repeated measurements under identical conditions, and the associated uncertainty mainly arises from spectral line fitting and signal fluctuations.

$$\ln \frac{I_{ij} \lambda_{ij} c_B}{g_i A_{ij}} = -\frac{E_i}{kT} + \ln \left[\frac{N_s}{U(T)} \right] \quad (1)$$

$$n_e = \frac{\Delta \lambda_{\text{Stark}}}{2 \times 10^{16} w} \quad (2)$$

3. Result and discussion

3.1. Laser-plasma interaction

The interaction mechanism of the non-scanning laser-arc is shown in Fig. 6. Due to the concentrated laser energy, a narrow and high laser plasma is formed above the keyhole, which contains a large number of ions and electrons. Under appropriate laser-arc separation conditions, the arc is attracted to the laser plasma and shifts towards the keyhole. Therefore, the plasma height above the keyhole is the highest. Since the continuous action of the laser makes the keyhole a continuous source of electron emission, a relatively stable cathode spot will be formed near the keyhole. According to the principle of minimum voltage, a parabolic conductive channel is formed between the cathode spot within the coverage of the arc plasma and the anode region at the end of the welding wire, and a strong exchange of particles between the laser and the arc is formed through this channel.

The arc plasma morphology characteristics of non-scanning laser-arc hybrid welding of aluminum alloy shown in Fig. 7 confirm the above-mentioned laser-arc interaction mechanism. At the time from $t = 0.7$ to $t = 1$ ms, the arc current reaches its maximum value. At this time, the arc shape is jointly affected by the plasma flow effect and the charge flow effect. The arc area is the largest, and there is obvious expansion in the height direction near the keyhole, making the arc shape appear as "bell-shaped". Subsequently, at the time of $t = 1.57$ ms, as the current decreases, the plasma flow effect weakens, causing the arc to contract. However, there is an obvious bulge in the middle position of the arc near the keyhole side, indicating that the laser plasma still has a strong attraction to the arc. The charge flow effect makes the conductive channel still established between the welding wire and the keyhole during the arc current decrease process and deviates from the welding wire axis in a parabolic shape.

The typical arc plasma morphology characteristics of different scanning modes and scanning parameters are shown in Fig. 8 ($A = 0.6$ mm, $f = 300$ Hz). At the peak current moment ($t = 1$ ms), the arc shape is "bell-shaped". During the current decreasing stage from $t = 1.14$ to $t =$

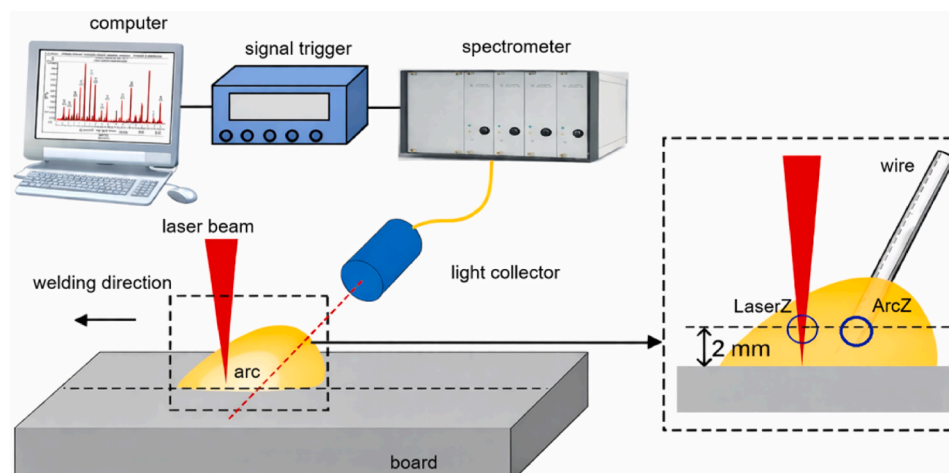


Fig. 4. Spectral signal acquisition platform and collection positions.

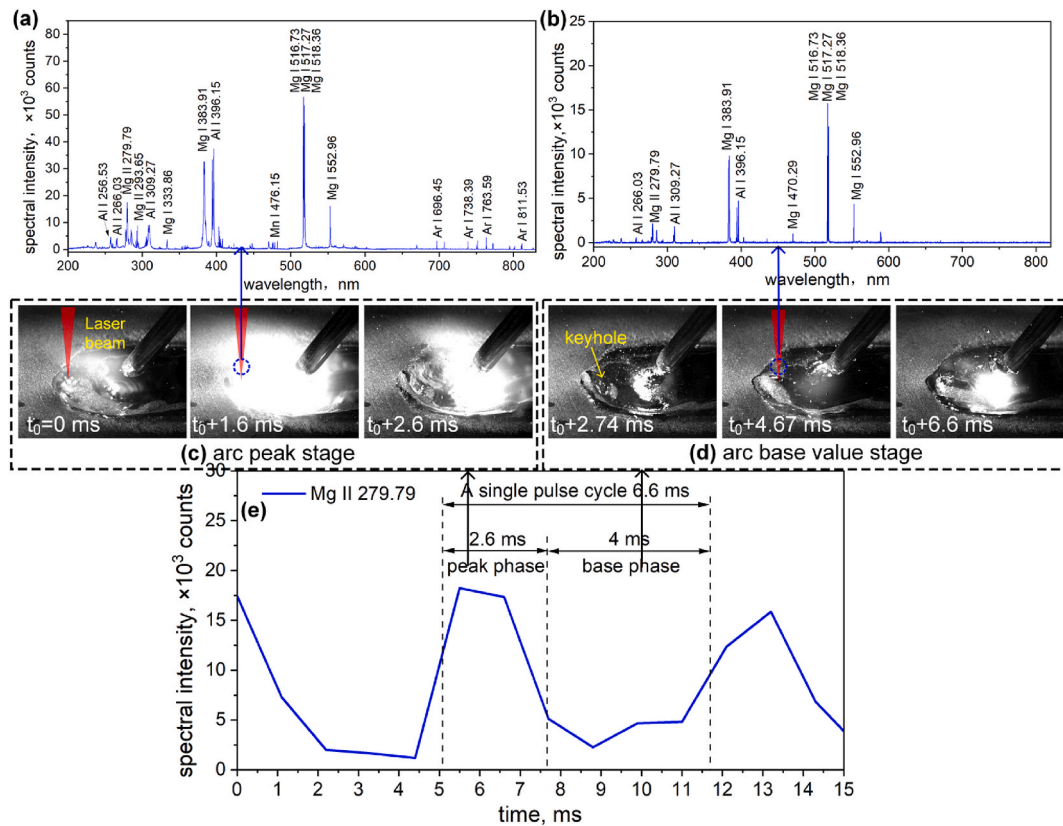


Fig. 5. Typical spectral characteristics in the laser irradiated zone (LaserZ) and high-speed photography results of the welding process for non-scanning laser-MIG hybrid welding of aluminum alloy (Laser power $P = 5000$ W, Welding current $I = 200$ A, Welding speed $v_w = 2$ m/min). (a)-(b) Spectral characteristics during the peak and base current phases of the arc; (c)-(d) High-speed photography results during the peak and base current phases of the arc; (e) Time-domain distribution of the characteristic spectral line Mg II 279.79 nm.

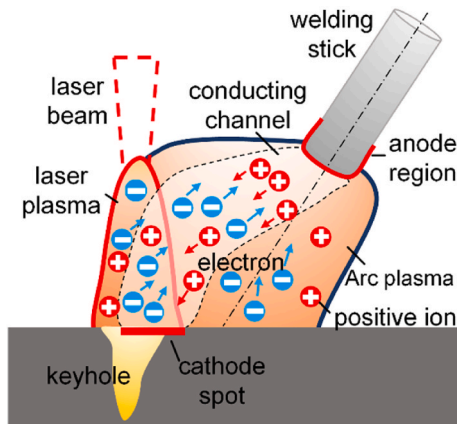


Fig. 6. The mechanism of laser-arc interaction under non-scanning (charge flow effect).

1.7 ms, the arc has a certain degree of contraction. Under the traction of the beam oscillating, the shape of the arc plasma changes regularly with the movement of the keyhole. For the transverse scanning mode, the end of the arc near the keyhole forms a bulge and reciprocates along the weld centerline. For the radial scanning mode, the bulging part of the arc is continuously stretched and compressed with the laser beam scanning, swinging back and forth along the radial direction. For the circular scanning mode, as the conductive channel swings, more arcs are introduced into the laser irradiation area. The arc remains relatively stable during the current decreasing stage, and the overall fluctuation is small. It can be seen that the circular scanning mode promotes the exchange of particles in the composite welding plasma and keeps the arc in a relatively stable shape during the peak stage.

Because the pulsed arc mode is used in this paper, the maximum current reaches 450A during the peak phase of the pulse current, while the arc base current is only 50A. Therefore, the interaction between the laser plasma and the arc plasma mainly occurs during the arc peak phase. To simplify the problem, assuming that the arc plasma volume is the same under different laser beam scanning modes, the laser plasma

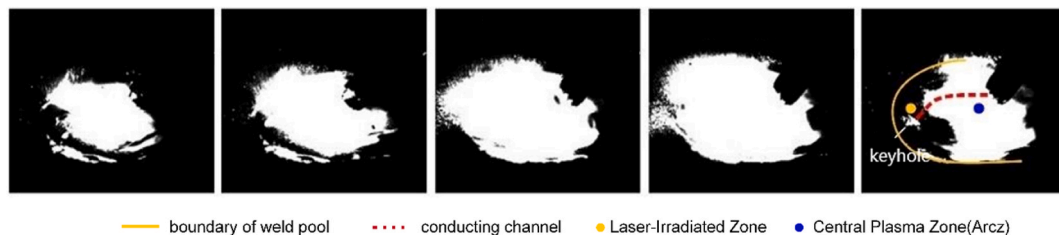


Fig. 7. High-speed photography image of the composite welding arc under non-scanning laser ($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min).

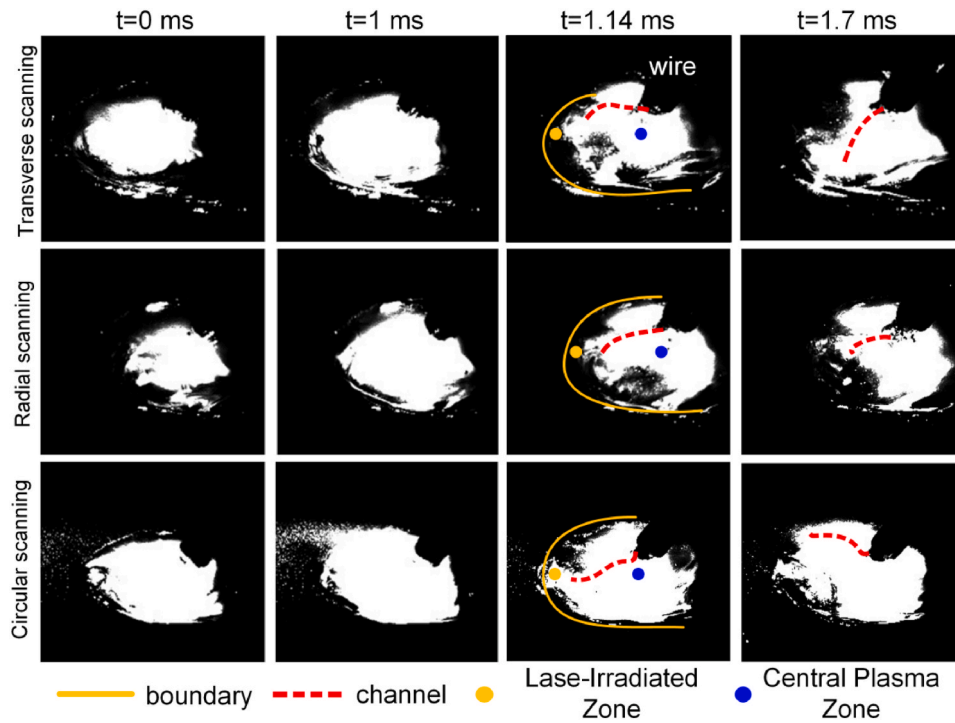


Fig. 8. High-speed photography comparison of arc plasma under different scanning modes ($A = 0.6$ mm, $f = 300$ Hz).

and the arc plasma are projected onto a two-dimensional plane, so that the complex three-dimensional spatial problem of the relative position of beam oscillating and arc is simplified to a two-dimensional spatial problem.

Taking the scanning amplitude $A = 0.6$ mm as an example, as shown in Fig. 9, the arc plasma intensity is the highest at the arc peak moment, and the arc plasma completely covers the laser plasma. At this time, the laser-arc interaction is the strongest. As the arc current decreases, the arc plasma gradually decreases. For the non-scanning mode, since the laser beam position is fixed, the laser plasma and arc plasma interaction weakens as the arc current decreases. Beam oscillating can increase the

area of the laser plasma and arc plasma interaction zone. Especially for the beam circular scanning, the interaction area between the laser plasma and the arc plasma it forms is larger than that of other laser beam scanning modes. During the scanning rotation process, the laser beam brings more electrons in the laser plasma into the ArcZ area through the conductive channel. At the same time, the positive ions in the arc will also enter the LaserZ area along with the conductive channel. Therefore, the particle exchange between the two is strengthened.

The degree of laser-arc interaction is quantitatively discussed below based on the spectral characteristics of the LaserZ and ArcZ zones.

As shown in Fig. 10(a)–(b), comparing the average spectral intensity changes of the LaserZ and ArcZ zones under different modes reveals that the spectral intensity in the LaserZ zone for non-scanning and transverse scanning modes is the lowest, around 6800 counts; the spectral intensity for the longitudinal scanning mode is relatively higher, reaching 7543 counts; the spectral intensity for the circular scanning mode increases significantly to 9125 counts, a 30% increase compared to the non-scanning mode. For the ArcZ zone, the spectral intensity for non-scanning is the lowest, at 20468 counts; the spectral intensities for the transverse and longitudinal scanning modes are similar, around 23500 counts, an increase of 14.6% compared to non-scanning; the spectral intensity for the circular scanning mode reaches 27366 counts, an increase of 33% compared to non-scanning. These data indicate that beam oscillation scanning, especially circular scanning, attracts more Mg II ions into the LaserZ zone and promotes the ionization of more Mg I ions in the ArcZ zone. This measurement result is consistent with the high-speed photography results of the arc plasma shown in Figs. 7 and 8, thus confirming the reliability of the spectral intensity data. Therefore, compared to the transverse and longitudinal scanning modes, the circular scanning mode enhances laser-arc interaction the most.

As shown in Fig. 11(a)–(b), the electron temperature in the LaserZ and ArcZ zones increases sequentially with the modes: non-scanning, transverse, longitudinal, and circular scanning. For the LaserZ zone, the electron temperature for circular scanning is approximately 13200 K, which is about 1100 K higher than non-scanning, an increase of about 9%. The electron temperatures for transverse and longitudinal scanning are 12606 K and 12745 K, respectively. For the ArcZ zone, the electron

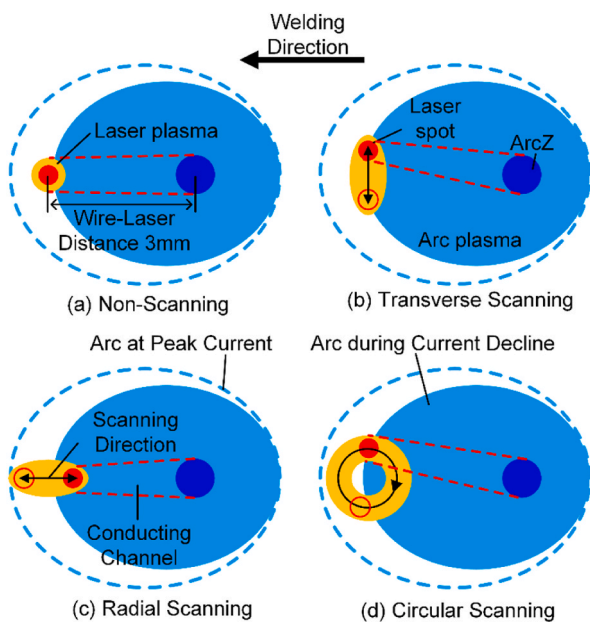


Fig. 9. The positional relationship between laser plasma and arc plasma under different laser beam scanning modes.

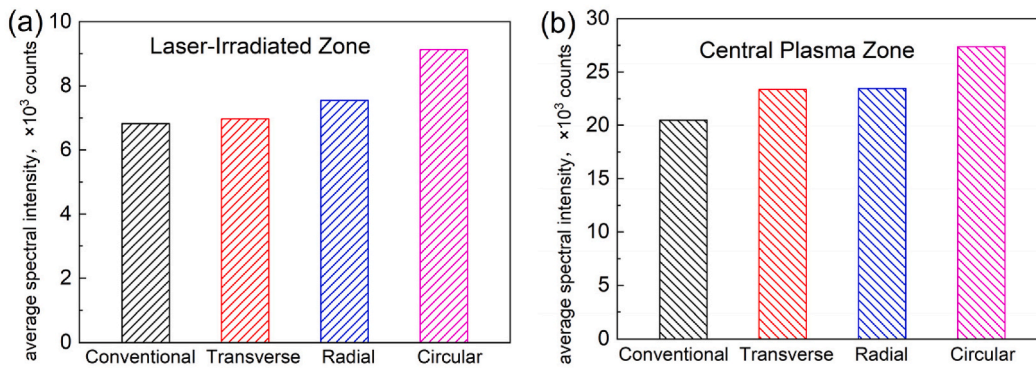


Fig. 10. The average spectral intensity under different scanning modes($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

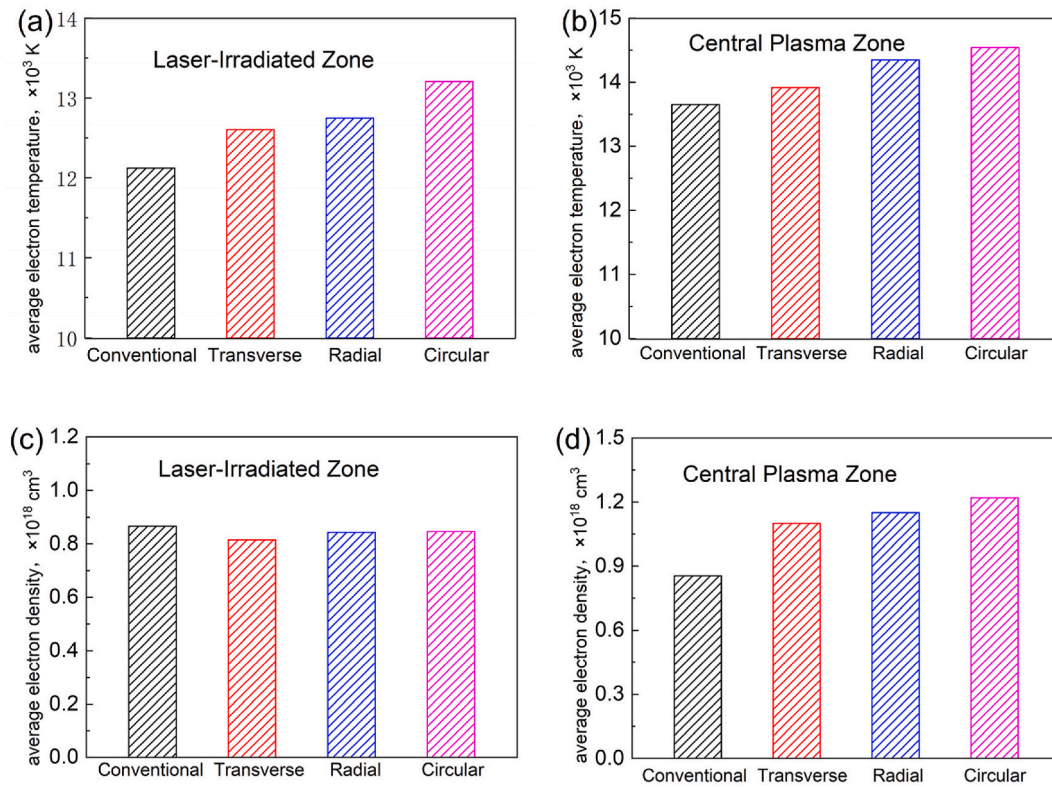


Fig. 11. The average electron temperature and density under different scanning modes($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

temperature for circular scanning reaches 14540 K, which is 1140 K higher than that of non-scanning, an increase of about 8.5%.

According to literature [20,21], the plasma resistivity can be expressed by Formula 3,

$$\eta = \frac{\pi e^2 m_e^{1/2}}{(4\pi\epsilon_0)^2 (\kappa T_e)^{3/2}} \ln \Lambda \quad (3)$$

where e is the electron charge, m_e is the electron mass, κ is the Boltzmann constant, T_e is the electron temperature, ϵ_0 is the permittivity, and $\ln \Lambda$ is the Coulomb logarithm. Since other parameters can be taken as constants, the plasma resistivity is a function of the electron temperature. Clearly, the higher the electron temperature, the lower the plasma resistivity. Therefore, compared to other scanning modes, circular scanning has better conductivity in both the LaserZ and ArcZ zones, which facilitates arc excitation and wire melting.

As shown in Fig. 11 (c), the electron density in the LaserZ zone differs little across scanning modes, approximately $(0.82\sim 0.86) \times 10^{18}$ cm $^{-3}$.

This is mainly because although laser beam scanning promotes particle exchange, the high energy density of the laser focal point is the main source of electron emission, with electrons primarily moving through the conductive channel to the anode region. Therefore, the number of electrons entering the LaserZ zone from the ArcZ zone is relatively small, leading to little change in electron density. However, laser beam scanning enhances the electron transport from the laser plasma to the ArcZ zone, making the electron density changes more noticeable in the ArcZ zone.

As shown in Fig. 11 (d), the electron density in the ArcZ zone for non-scanning is the lowest, approximately 0.85×10^{18} cm $^{-3}$. Under laser beam scanning, the electron density in the ArcZ zone is significantly higher than in non-scanning, with the circular scanning mode having the highest electron density at 1.22×10^{18} cm $^{-3}$, a 43% increase compared to non-scanning. The increase in electron density clearly helps to enhance the conductivity of the ArcZ zone, consistent with the previously detected electron temperature results.

3.2. Effect of scanning mode on droplet transfer

3.2.1. Effect of scanning mode on droplet transfer

As shown in Fig. 12, according to Danut's summary of droplet transfer modes in gas metal arc welding, for small and medium currents (<200A), droplet transfer is usually dominated by short-circuiting transfer, globular transfer (droplet diameter much larger than welding wire diameter and no obvious liquid metal column formation), and repelled transfer. For medium currents (150–250A), droplet transfer usually exhibits a spray transfer mode, and according to droplet size, it can be divided into coarse projected transfer (droplet diameter larger than welding wire diameter), medium projected transfer (droplet diameter slightly larger than or equal to welding wire diameter), and fine projected transfer (droplet diameter smaller than welding wire diameter). Medium projected transfer and fine projected transfer were mainly distinguished based on the droplet diameter and detachment frequency, with fine projected transfer characterized by smaller droplet size and higher transfer frequency. For large currents (>250A), droplet transfer is mainly dominated by spray transfer and rotating spray transfer modes [22]. The following will be used as a basis for judging the droplet transfer mode.

It can be seen from Fig. 13 that for the non-scanning mode, it takes a total of 11.25 ms for the droplet to form and enter the molten pool to complete the transfer, which experiences two arc pulse peak stages ($t = 2.29$ ms and $t = 8.39$ ms). The droplet transfer process can basically be decomposed into the following four stages: droplet formation, droplet detachment, droplet transfer, and falling into the molten pool. In the droplet formation stage ($t = 0$ to $t = 2.86$ ms), the welding wire end melts rapidly under the high current of the peak arc, and the droplet begins to form. Under the action of the arc contraction force, the droplet grows into a liquid metal column and has a certain degree of necking. The necking diameter D_R is about 0.82 mm. The growth direction of the liquid column points to the molten pool along the welding wire centerline, but the arc energy of the first pulse peak is obviously not enough to make the droplet detach from the welding wire. In the droplet detachment stage ($t = 2.86$ to $t = 8.39$ ms), the droplet detachment process can be divided into two steps. At the time from $t = 6$ to $t = 7.53$ ms, the top of the liquid metal column tends to be spherical under the action of surface tension, and the droplet diameter exceeds the welding wire diameter, reaching 1.85 mm. At this time, the coarse droplet does not continue to transfer to the molten pool along the welding wire centerline, but raise to the left front. When the second arc peak current is formed ($t = 8.39$ ms), the droplet starts to detach from the welding wire under the action of the arc force. In the droplet transfer stage ($t = 9.96$ ms), the falling direction of the droplet deviates from the welding wire centerline and falls towards the position close to the keyhole. In the droplet falling into the molten pool stage ($t = 11.25$ ms), the droplet falls into the molten pool, causing violent fluctuations in the leading edge of the molten pool and generating coarse spatter. It can be seen that the non-scanning mode requires the droplet to grow to a size close to or larger than the welding wire diameter to effectively transfer to the molten pool. This is a typical coarse projected transfer mode. This

transfer mode usually occurs under the laser-arc hybrid welding conditions of small and medium currents (<200A). Cai et al. also made similar findings in the study of droplet transfer in laser-arc hybrid welding of carbon steel [23]. Under this droplet transfer mode, since the droplet volume is large, it has a large momentum when falling, which easily causes molten pool oscillation and forms coarse spatter when entering the molten pool. In addition, in this mode, the droplet needs to go through two or more arc pulse cycles to transfer to the molten pool. The droplet transfer time needs at least 9 ms or more, and the transfer time is long. The fish scale pattern formed on the weld surface is coarse and not dense (Fig. 3(a)).

In addition to the coarse projected transfer mode, the non-scanning droplet transfer also has a short-circuiting transfer mode, as shown in Fig. 14. When the arc peak stage just ends ($t = 2.8$ ms), the droplet grows and lengthens, and the droplet diameter is 1.6 mm. As the droplet continues to grow to form a liquid metal column with a diameter of about 0.89 mm, the droplet is difficult to detach until it contacts the molten pool surface ($t = 3.1$ ms), and the arc is short-circuited and extinguished. At the time of $t = 3.4$ ms, the droplet overheats and vaporizes, causing an explosion and forming a large amount of spatter. The droplet completes the transfer within an arc pulse cycle, and the transfer takes about 4 ms.

As shown in Figs. 15 and 16, the droplet transfer under the transverse scanning mode is mainly dominated by medium projected transfer and short-circuiting transfer modes. During the laser beam scanning process, the real-time position and movement direction of the keyhole at the corresponding moment located on the scanning path are marked in the lower right corner of each picture. As shown in Fig. 15 at the time of $t = 3.15$ ms, compared to the non-scanning mode, the liquid bridge necking diameter D_R of the transverse scanning is reduced to about 0.6 mm. As shown in $t = 3.72$ ms in the figure, the droplet detaches from the liquid column and enters the molten pool at $t = 4$ ms. The droplet diameter is about 1.7 mm when detaching, which is slightly larger than the welding wire diameter, indicating that the transfer mode at this time is an medium projected transfer mode. It is worth noting that at this time, the keyhole moves to the edge of the oscillating path, and the falling direction of the droplet points to the position where the keyhole is located, indicating that the keyhole has a strong attraction to the droplet and changes the falling trajectory of the droplet. From $t = 4.29$ to $t = 5.15$ ms, the remaining liquid metal column forms small-particle droplets and falls into the molten pool. The droplet transfer is basically one droplet per pulse, and the transfer time is greater than 4 ms. As shown in Fig. 16 at the time of $t = 3.27$ ms, when the keyhole moves to the center of the scanning path, the droplet happens to transfer to the molten pool and contacts the molten pool before detaching, causing a short circuit and forming a large amount of spatter, indicating that the transfer mode is short-circuiting transfer. In this scanning mode, fine spatter zones are formed on both sides of the weld surface (Fig. 3(b)).

The radial scanning mode is dominated by medium projected transfer and short-circuiting transfer modes. Among them, for the projected transfer mode, the droplet often shows the characteristics of swinging along the radial direction during the transfer process. As

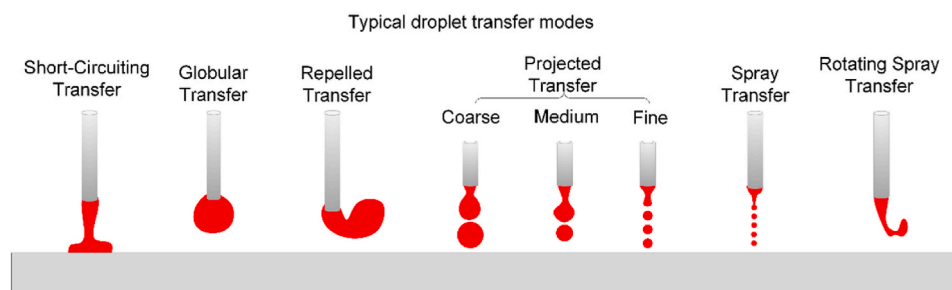


Fig. 12. Typical droplet transfer mode.

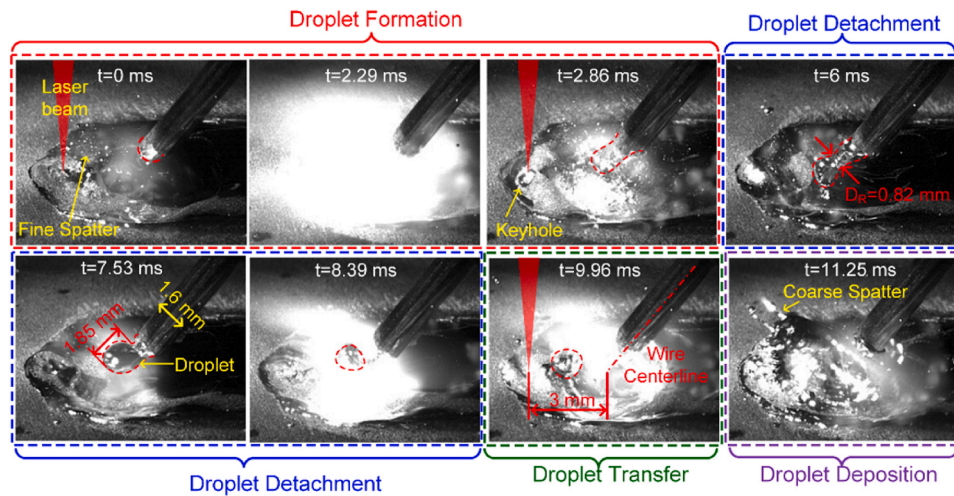


Fig. 13. Non-scanning laser-arc composite welding droplet transition behavior (coarse projected transfer)($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

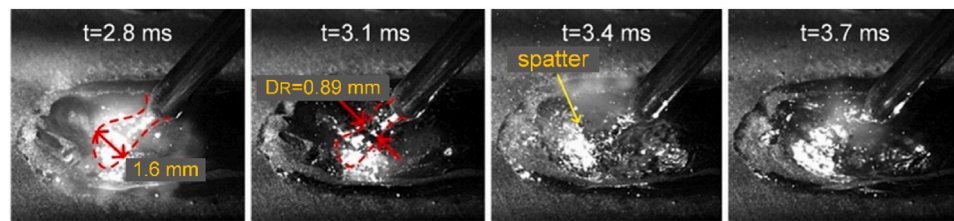


Fig. 14. Non-scanning laser-arc composite welding droplet transition behavior (short-circuiting transfer)($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

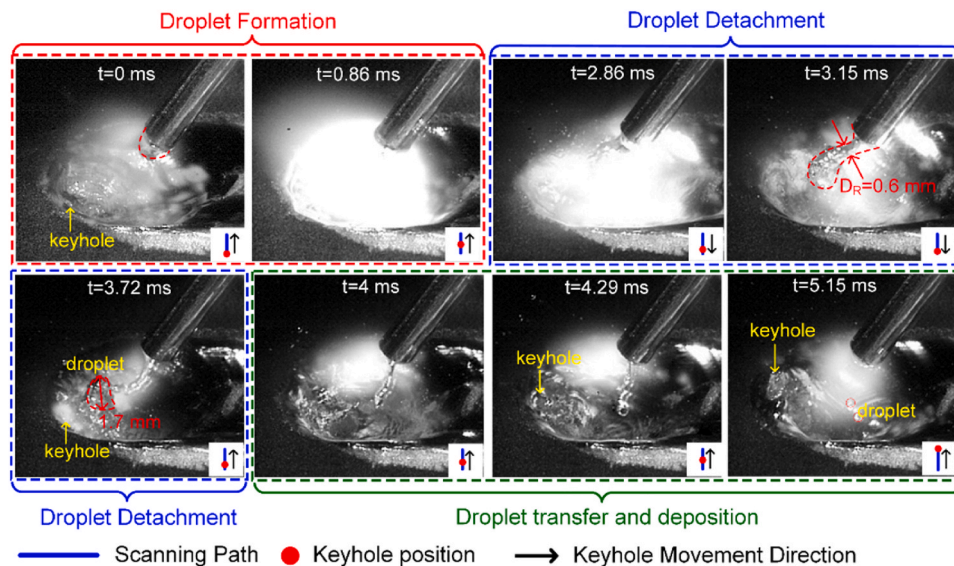


Fig. 15. Transverse scanning laser-arc composite welding droplet transition behavior (medium projected transfer)($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

shown in Fig. 17, at the time from $t = 2.57$ to $t = 3$ ms, the laser beam moves to the side of the scanning path close to the droplet. At this time, the laser beam directly irradiates the droplet, causing the droplet to overheat and expand and separate a part of the droplet with a diameter of 1.15 mm and fall into the molten pool. At the time from $t = 3$ to $t = 3.43$ ms, the keyhole moves away from the droplet. At this time, the remaining liquid metal column swings towards the back of the molten

pool. At the time from $t = 4.55$ to $t = 6.6$ ms, the droplet with a diameter of about 0.8 mm detaches from the liquid column and flies towards the back of the molten pool. This spray transfer mode with reciprocating swing detachment characteristics makes it easy to form coarse spatter on the weld surface. For the short-circuiting transfer mode of the droplet, as shown in Fig. 18, at the time from $t = 0.14$ to $t = 0.57$ ms, the laser irradiates the droplet surface, causing it to heat up and expand and

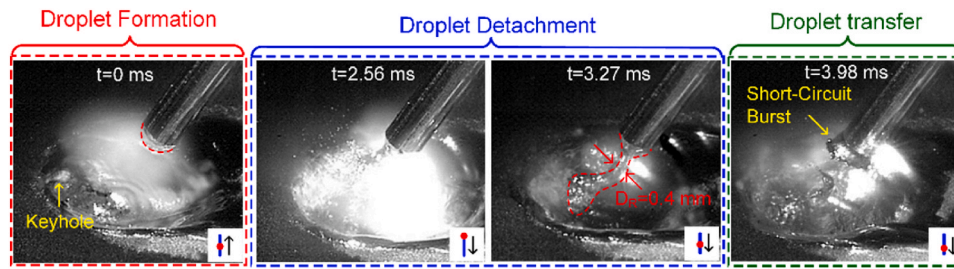


Fig. 16. Transverse scanning laser-arc composite welding droplet transition behavior (short-circuiting transfer) ($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

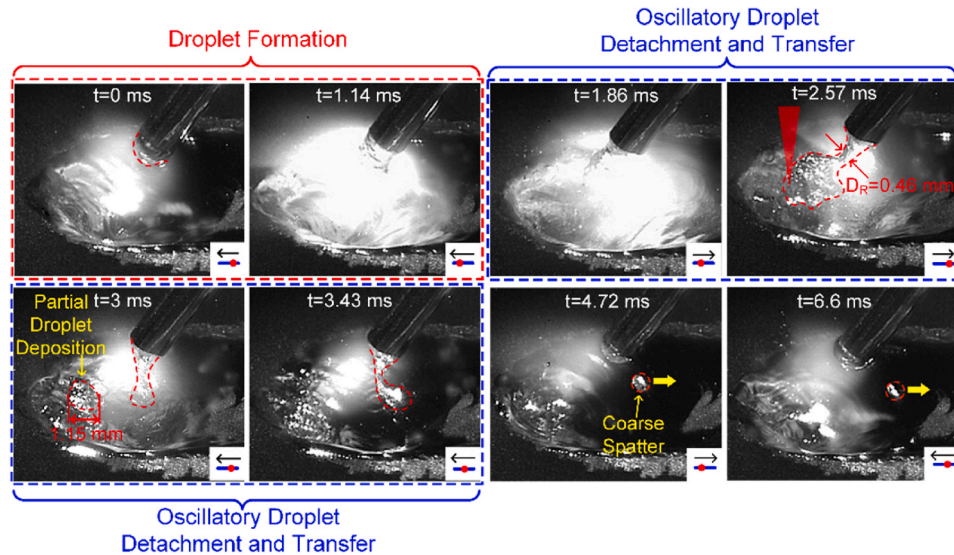


Fig. 17. Radial scanning laser-arc composite welding droplet transition behavior (oscillating projected transfer) ($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

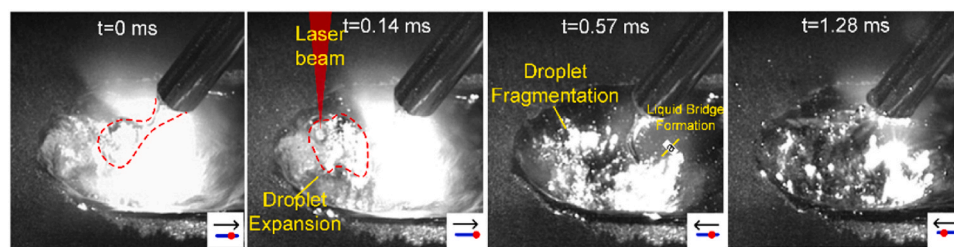


Fig. 18. Radial scanning laser-arc composite welding droplet transition behavior (short-circuiting transfer) ($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

explode and break. The remaining liquid metal column is too close to the molten pool. After contacting the molten pool, a short circuit occurs, causing a large amount of spatter. In this scanning mode, there are still coarse spatter on the weld surface (Fig. 3(c)).

As shown in Fig. 19, the droplet transfer of circular scanning laser-arc hybrid welding exhibits a fine projected transfer mode. As shown at the time of $t = 3.28$ ms, the necking diameter D_R of the liquid metal column formed by the droplet is 0.49 mm. At $t = 3.43$ ms, the liquid metal column is further necked. At $t = 3.74$ ms, the first droplet begins to transfer, and the droplet diameter is about 0.9 mm. At the time of $t = 4.15$ ms, the remaining liquid metal column forms a smaller-diameter droplet and transfers to the molten pool, which is a typical fine projected transfer mode. It can be seen that the spatter caused during the droplet transfer process is extremely small. In this scanning mode, the weld surface exhibits a fine and dense fish-scale pattern, and there are

almost no coarse spatter (Fig. 3(d)).

7000 high-speed photography photos within a welding time of 1 s were selected to statistically analyze the average droplet transfer time and average droplet diameter under different scanning modes, as shown in Fig. 20. It can be seen from this that the non-scanning mode has the longest average droplet transfer time and the largest average droplet diameter, which are 6.8 ms and 1.7 mm, respectively. The average droplet transfer times of transverse and radial scanning are similar, about 4.6 ms, which is nearly 32% smaller than that of the non-scanning mode. The circular scanning has the shortest average droplet transfer time and the smallest average droplet diameter, which are 3.8 ms and 0.9 mm, respectively. Compared with the non-scanning mode, the average droplet transfer time and droplet diameter are reduced by 44% and 47%, respectively. The above high-speed photography observation results of droplet transfer show that adding beam oscillating in laser-arc

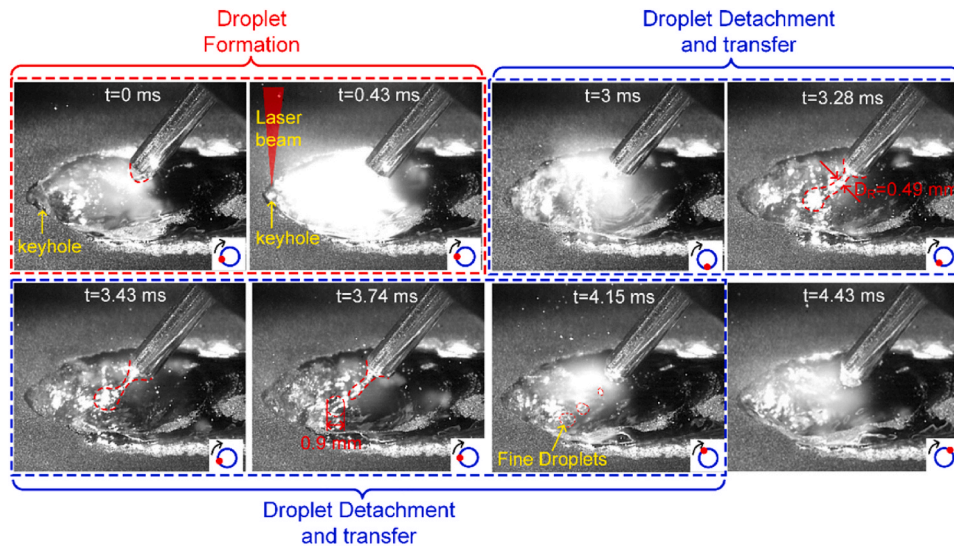


Fig. 19. Circular scanning laser-arc composite welding droplet transition behavior (fine projected transfer) ($P = 5000$ W, $I = 200$ A, $v_w = 2$ m/min, $A = 0.6$ mm, $f = 300$ Hz).

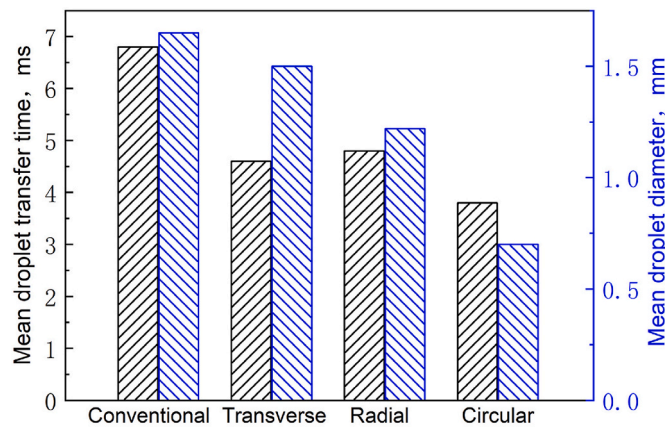


Fig. 20. Comparison of mean droplet transition time and mean diameter under different laser beam scanning modes.

hybrid welding is conducive to accelerating droplet detachment, especially circular scanning has the most obvious promotion effect on droplet transfer.

3.2.2. Physical mechanism of the effect of scanning mode on droplet transfer

From the results of the interaction between the laser and the arc plasma mentioned earlier, it can be known that the oscillating of the beam can increase the area of the interaction zone between the laser and the arc, promote the particle exchange between the two, and increase the melting rate and the amount of melting at the end of the welding wire. Since the welding current is fixed, that is, the wire feeding speed is constant, the length of the molten metal column is correspondingly longer, resulting in a change in the liquid metal column necking diameter D_R . The necking diameter D_R of the liquid metal column under different scanning modes is statistically analyzed and averaged, as shown in Fig. 21. The reported droplet diameter and transfer time values represent arithmetic mean values calculated from all valid measurements, and no outliers were intentionally excluded from the analysis. It can be seen from this that compared with non-scanning, the necking diameter of the liquid metal column scanned by the laser beam is significantly reduced. The radial and circular scanning are the smallest, about 0.45 mm, which is nearly half smaller than that of non-scanning.

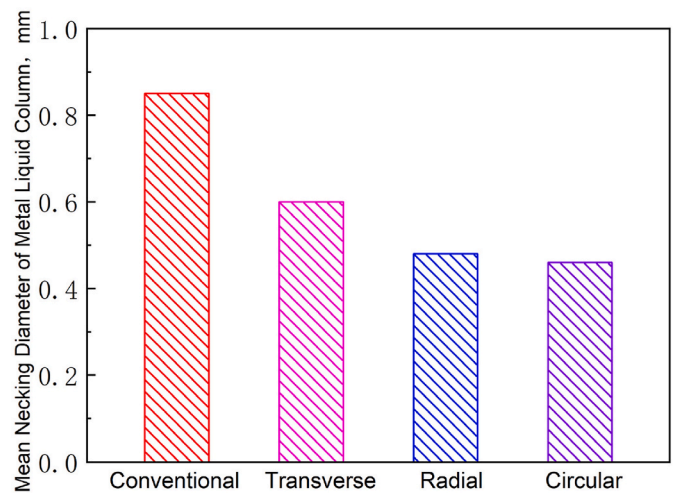


Fig. 21. Mean necking diameter of the molten metal column under different laser beam scanning modes.

The relationship between the liquid column diameter D_R and the droplet size can be explained based on the liquid stream instability theory [24]. The liquid metal column formed at the welding wire end is simplified as a cylinder. When it is disturbed, it will undergo the deformation shown in Fig. 22, where R is the initial radius of the liquid metal column, $R = D_R/2$; λ is the wavelength of the liquid column fluctuation. Formula 4 is the expression of the critical wavelength λ_c , where μ_0 is the vacuum permeability and γ is the surface tension. Its meaning is that when the liquid metal column disturbance grows to be

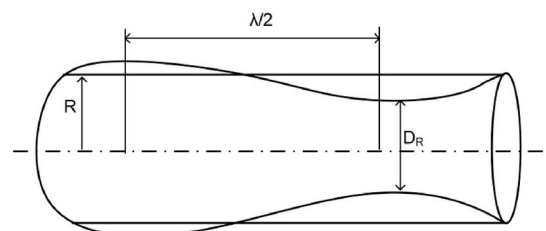


Fig. 22. Model of compression instability of the liquid flow beam.

greater than the critical wavelength λ_c , the droplet detaches from the liquid column. It can be seen that the smaller the necking D_R of the liquid metal column, the smaller λ_c , indicating that the liquid metal column is more unstable and the droplet is easier to detach. Therefore, the smaller the diameter of the droplet formed. It can be seen from Figs. 20 and 21 that the laser beam scanning reduces the necking diameter of the liquid metal column, so the droplet transfer diameter is reduced accordingly. This theory is consistent with the experimental phenomenon.

$$\lambda_c = \frac{2\pi R}{\left(1 + \frac{\mu_0 I^2}{2\pi\gamma R}\right)^{\frac{1}{2}}} = \frac{2\pi}{\left(\frac{4}{D_R^2} + \frac{8\mu_0 I^2}{2\pi^2\gamma D_R^3}\right)^{\frac{1}{2}}} \quad (4)$$

The previous high-speed photography experiment observed that the beam oscillating made the droplet transfer trajectory change significantly. This is mainly because the beam oscillating changed the force state of the droplet. In the non-scanning laser-arc hybrid welding process, the forces usually applied to the droplet include gravity F_g , arc mechanical force F_p (including plasma flow force and electromagnetic force), surface tension F_s , and metal vapor reaction force F_{RL} of the keyhole. The non-scanning droplet transfer force model is shown in Fig. 23.

The gravity applied to the droplet can be expressed by Formula 5,

$$F_g = \frac{4}{3}\pi R_D^3 \rho_d g \quad (5)$$

Where R_D is the droplet radius, which is 0.6 mm in this paper. ρ_d is the droplet density, and the angle between it and the welding wire centerline is defined as β . The welding wire used is ER5087 welding wire, and its density is $2.7 \times 10^3 \text{ kg/m}^3$. g is the gravitational acceleration, and the value is 9.8 m/s^2 . Gravity promotes droplet transfer. According to the droplet diameter under different scanning modes in Fig. 15, the corresponding droplet gravity can be calculated, which are $2.17 \times 10^{-5} \text{ N}$, $1.2 \times 10^{-5} \text{ N}$, $0.76 \times 10^{-5} \text{ N}$ and $0.15 \times 10^{-5} \text{ N}$ for non-scanning, transverse scanning, radial scanning, and circular scanning, respectively.

The arc mechanical force F_p applied to the droplet mainly includes plasma flow force and electromagnetic contraction force. It is generally believed that it is proportional to the square of the welding current, as shown in Formula 6,

$$F_p = KI^2 \quad (6)$$

where I is the arc current, K is an empirical constant, and it can be known from the literature [25] that the value is between $(1.2 \sim 1.4) \times 10^{-6} \text{ N/A}^2$. According to the current value of 200A selected in this paper,

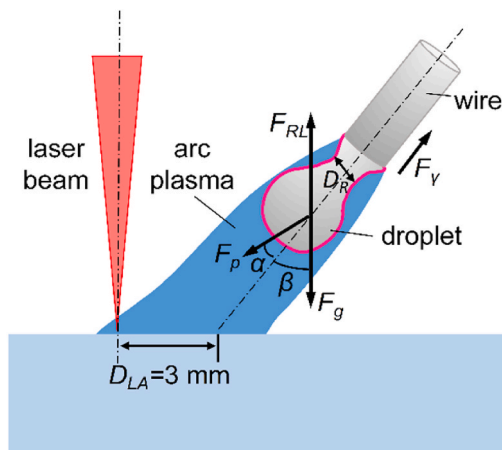


Fig. 23. Force model of droplet transition without scanning laser-electric arc composite welding.

K is $1.35 \times 10^{-6} \text{ N/A}^2$. Therefore, it can be obtained from the formula that the arc mechanical force is about 0.054 N. Under the attraction of the laser keyhole, it points to the position where the laser keyhole is located, and the angle with the welding wire centerline is α .

The liquid droplet is adhered to the welding wire end by the surface tension F_s , which hinders the droplet transfer. This force can be expressed by Formula 7,

$$F_s = 2\pi R_w \sigma \quad (7)$$

where R_w is the welding wire radius, which is 0.8 mm in this paper; σ is the surface tension coefficient, which is related to the material composition, temperature, and shielding gas, and can be taken as 0.9 N/m in this paper [26]. Therefore, it can be calculated that $F_s = 4.52 \times 10^{-3} \text{ N}$. In the laser-arc hybrid welding process, the keyhole formed by the laser will eject a large amount of metal vapor, which hinders the droplet. Assuming that the laser heat source is a Gaussian rotating body heat source, and the radius of its action area is $r_0(z)$, which decays regularly along the thickness direction of the welded material, the metal vapor force caused by the laser keyhole can be expressed by Formula 8 [6],

$$F_{RL} = \frac{1}{2} C_D A_p \rho_m^2 V_0^2 \left(\frac{N_a k_B T_s^{3/2}}{M_a B_0} \right) \exp\left(-\frac{U}{T_s}\right) \exp\left(-\frac{3r^2}{r_0^2(z)}\right) \quad (8)$$

where V_0 is a constant, M_a is the gas molecular weight, N_a is Avogadro's number, k_B is Boltzmann's constant, and T_s is the surface temperature of the molten metal. U is the calculated value, $U = M_a L_v / N_a k_B$; L_v is the latent heat of vaporization of the melt; B_0 is the evaporation constant of the melt. According to the calculation results of metal vapor recoil pressure by Xu et al., when the laser-arc separation distance is greater than 1.5 mm, the metal vapor recoil pressure has little effect on droplet transfer and can be ignored [27]. According to the above force analysis of the droplet, the arc mechanical force is one to two orders of magnitude larger than other forces. Therefore, the droplet transfer is mainly determined by the arc mechanical force.

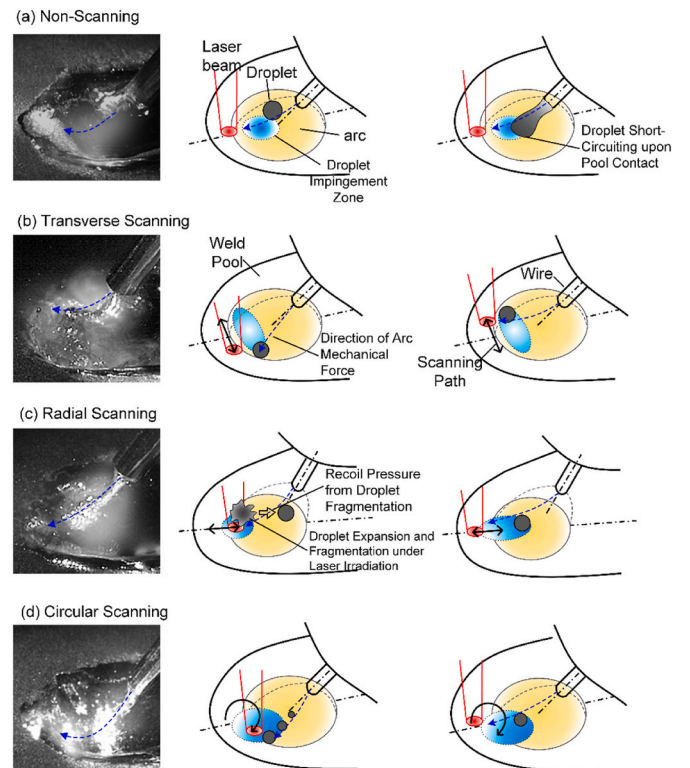


Fig. 24. Droplet transition mechanism under different laser beam scanning modes.

Based on the above laser-arc interaction and droplet force analysis, Fig. 24 illustrates the influence mechanism of different scanning modes on droplet transfer behavior.

As shown in Fig. 24(a), for the non-scanning mode, the relative position of the keyhole and the welding wire end is fixed. The direction of the arc mechanical force always points to the keyhole, and the droplet falls into the molten pool behind the keyhole along the direction of the arc force. Because the necking diameter of the liquid metal column is large, it is difficult for the droplet to detach in the non-scanning mode. Therefore, it is easy to contact the molten pool before detaching and cause a short circuit, resulting in a large amount of spatter and affecting process stability.

As shown in Fig. 24(b), for the transverse scanning mode, the keyhole reciprocates along the direction perpendicular to the molten pool centerline. At the turning point of the transverse scanning path, the laser beam scanning speed is the smallest, and the interaction between the laser and the molten pool is the strongest. At this time, the laser plasma above the keyhole has the strongest attraction to the arc, making the droplet fall into positions biased towards both sides of the molten pool, increasing the probability of spatter on both sides of the molten pool. Therefore, for the transverse scanning mode, spatter zones are easily formed on both sides of the weld, affecting the weld surface quality.

As shown in Fig. 24(c), for the radial scanning mode, the keyhole reciprocates along the molten pool centerline, and the relative position of the keyhole and the welding wire end is sometimes short and sometimes long. When the keyhole moves to the turning point position close to the welding wire end, there is a greater probability that the laser beam will irradiate the droplet surface, causing the droplet to overheat and expand and explode. While generating a large amount of spatter, it forms a large recoil pressure on the remaining liquid metal column, causing the liquid metal column to swing towards the back of the molten pool, increasing the probability of the formed droplet falling into the back of the molten pool. If the droplet falls on the solidified or rapidly solidifying weld surface, it will adhere to the weld surface and form spatter.

As shown in Fig. 24(d), for the circular scanning mode, the keyhole moves clockwise along the scanning path, and the arc mechanical force also swings clockwise with the keyhole. When the keyhole moves to a position away from the welding wire end, the attraction of the keyhole to the droplet weakens. At this time, the droplet falls into a position close to the back of the keyhole on the molten pool centerline. Since the circular scanning process does not have turning points similar to the transverse and radial scanning modes, and the circular scanning mode forms the smallest liquid metal column necking diameter, the droplets formed are also the smallest and easier to detach from the liquid metal column. The probability of the beam directly irradiating the droplet during the scanning process is small, making the circular scanning mode droplet transfer smoother and the spatter smaller. The above analysis shows that compared with the beam transverse and radial scanning modes, the beam circular scanning is conducive to improving process stability, enhancing laser-arc interaction, and improving droplet transfer characteristics.

Therefore, the oscillating laser scanning enhances the interaction between the laser-induced plasma and the arc plasma, resulting in a more stable arc and intensified electromagnetic forces acting on the molten droplet. As a result, droplet detachment is promoted and refined projected transfer behavior is achieved.

4. Conclusion

- (1) Oscillating laser scanning, particularly the circular mode, significantly enhances laser-arc plasma interaction. Circular scanning increased the spectral intensity of the characteristic Mg II line by 30% in the laser zone and 33% in the arc zone compared to non-scanning. It also raised the arc zone electron temperature

by ~8.5% and electron density by 43%, effectively reducing arc resistivity and improving arc stability and heating efficiency.

- (2) The enhanced plasma interaction directly improves droplet transfer. Circular scanning uniquely promotes stable fine projected transfer, reducing the average droplet diameter to 0.9 mm and transfer time to 3.8 ms—47% and 44% reductions, respectively, compared to non-scanning—while drastically suppressing spatter. Transverse and longitudinal scanning mainly yield mixed modes of medium projected and short-circuiting transfer with higher spatter tendency.
- (3) The physical mechanism involves a synergistic effect: The improved arc conductivity promotes wire melting, forming a thinner liquid metal column (necking diameter $D \sim R \sim$ reduced to ~ 0.45 mm). According to instability theory, this smaller $D \sim R \sim$ decreases the critical wavelength for breakup, facilitating droplet refinement. Furthermore, the uniform circular motion provides a dynamically stable arc force direction, avoiding disturbances and guiding fine droplets smoothly into the weld pool.
- (4) Circular scanning demonstrates clear process advantages for high-quality aluminum alloy laser-arc hybrid welding. It stabilizes the arc, refines droplet transfer, minimizes spatter, and improves weld surface appearance. The optimal processing window identified is circular scanning with an amplitude between 0.4 and 1.0 mm and a frequency of 200 Hz or higher.

Although the present study focuses on AA6082 aluminum alloy, the proposed mechanism related to enhanced laser-arc plasma interaction and droplet refinement is expected to be qualitatively applicable to other aluminum alloys under similar laser-arc hybrid welding conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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